

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I finish an M.Sc. in Economics at the University of Cologne this month, with a specialisation in energy economics and a thesis in applied microeconometrics, and I am applying for the doctoral position on auction-based support schemes for renewable energy. Energy economics has followed me through both degrees. What kept me in it is an argument I first met in Dietmar Lindenberger's module on growth, energy and climate change, that energy is the capacity to do work, including the work that activates capital, so its contribution to output runs ahead of its cost share. If that holds, energy is a production factor we have not finished internalising, and the rules that price it and pay for it decide whether prosperity and a cleaner path than our current one are available together. I want to spend four years on how the design of a support scheme changes what actually gets built, and I want to do that at CORE.

CORE describes itself as an interdisciplinary centre that brings operations research, economics and econometrics together, and it names the training of doctoral researchers as one of its objectives. That is the reason I am writing to you and not to a department that does one of those three things. My own work sits on the econometrics side. What I do not yet have is graduate industrial organization and game theory at the level this project runs on, and a centre founded in 1966 on that tradition, with more than twenty professors and thirty-five doctoral students in one building, is where I would learn it properly instead of reading around it. Everything I have written so far I wrote alone, and four years inside a group that argues these questions formally is the part I want most. I have started on the theory without being asked to. I am working through Jehle and Reny in my own time, because I would rather arrive able to follow the seminar than be brought up to it.

What I would bring is the empirical half of your question. My thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, asked whether a national scholarship programme in Brazil raised formal-sector wages and for whom. I built the panel from administrative records, five million linked employer-employee observations across 558 microregions from 2005 to 2019, in R through BigQuery, and estimated Callaway and Sant'Anna (2021) group-time effects with doubly-robust inference across exposure bins. Low-exposure regions show a wage gain of 3.04 per cent, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response that two-way fixed effects reports with the opposite sign. That skill transfers here directly. Since the 2016 state aid framework member states have adopted technology-neutral, technology-specific and mixed auction designs at different dates, which is staggered adoption of a design rather than of a policy, and the AURES II database records the design and outcome of every renewable auction in the EU and UK from 2011 to 2021. I would estimate how joint bidding shifts the awarded technology portfolio and the rent per megawatt hour against not-yet-treated member states. Alongside the econometrics I have advanced microeconomics on strategic thinking and competition with Axel Ockenfels at 1.3, whose own field is market and auction design, 1.0 in mathematics, 1.3 in econometrics, and I have calibrated a model in MATLAB. I am free from October 2026, which I know matters for a post starting this autumn, and my French is B1.

Thank you for considering my application. Prof. Dr. Pia Pinger, who chaired my thesis, and Jun.-Prof. Dr. Oliver Ruhnau, who taught and examined my module on energy markets and regulation, have both agreed to act as referees. They can be reached at pia.pinger@uni-koeln.de and ruhnau@wiso.uni-koeln.de. I would be glad to talk the project through whenever it suits you.

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